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[22AS51]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

FLIGHT MECHANICS – I (PERFORMANCE)

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Classify aircraft on the basis of their wing position (with a rough sketch for each case).	L2	CO2	**02**
	b.	Outline the working principle of spoilers.	L2	CO3	**02**
	c.	Sketch boundary layer profile over a flat plate and identify laminar, transition and turbulent zone.	L2	CO4	**02**
	d.	Construct thrust available and thrust required versus velocity curve for a typical fighter jet at sea level.	L4	CO5	**02**
	e.	Compare the effect of Mach number on different types of turbomachines.	L4	CO6	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Sketch the variation of lift coefficient with deflection of slats and flaps.	L2	CO1	**05**
		OR			
	b.	Illustrate the position of high-lift devices during different maneuvers.	L2	CO1	**05**
3.	**a.**	Estimate profile drag of typical fighter aircraft components using boundary-layer theory.	L3	CO2	**05**
		OR			
	b.	Review the salient features of panel method for the determination of aerodynamic loads.	L3	CO2	**05**
4.	**a.**	Determine minimum and maximum speed during steady and level flight of a jet powered aircraft.	L4	CO3	**05**
		OR			
	b.	A glider having W = 2000 N, 5 = 8.0 m^2, A = 16.0, e = 0.95, and C_{DO} = 0.015 is launched from a height of 300 m. Determine the maximum range at sea level.	L4	CO3	**05**
5.	**a.**	Review the variation of resultant accelerating force during take-off ground run of a propeller aircraft.	L4	CO4	**05**
		OR			

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	b.	For a propeller powered aircraft, examine the following expression for maximum rate of climb. $$(V_{R/C})_{max} = \frac{1}{3^{1/4}}\sqrt{\frac{2W}{\rho S}}\left(\frac{k}{C_{D0}}\right)^{1/4}$$	L4	CO4	**05**
6.	**a.**	Investigate the condition of rolling stability of a fighter aircraft.	L4	CO5	**05**
		OR			
	b.	Investigate the condition of longitudinal static stability of a transport aircraft.	L4	CO5	**05**
7.	**a.**	Review mission schedule of short-range commercial jet airliner.	L4	CO6	**05**
		OR			
	b.	Review the payload-range performance of typical fighter aircraft.	L4	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Examine the aerodynamic characteristics of highly swept delta configurations at subsonic speeds.	L4	CO1	**10**
		OR			
	b.	Compare the effect of aspect ratio on the wing lift coefficient.	L4	CO1	**10**
9.	**a.**	Sketch the trailing edge vortex characteristics behind a conventional large aircraft.	L4	CO2	**10**
		OR			
	b.	Compare the effect of Mach number on different types of turbomachines.	L4	CO2	**10**
10.	**a.**	A turbojet airplane weighs 45,000 N and has a wing loading *W/S* of 1450 N/m^2. The drag polar is given by $C_D = 0.014 + 0.038C_L^2$, $C_{L,max} = 1.5$, and the thrust $T = 20{,}000$. Investigate the absolute ceiling of this airplane.	L4	CO3	**10**
		OR			
	b.	Assuming minimum power condition, derive an expression for the following: i. Lift-coefficient. ii. Velocity. Also, find a relationship with the same under minimum drag condition.	L4	CO3	**10**
11.	**a.**	The landing operation is divided into three parts – approach (s1), flare(s2) and ground run(s3). Investigate the ground run distance (s3).	L4	CO4	**10**
		OR			

	b.	Examine the condition under which an aircraft is likely to abort a take-off.	L4	CO4	**10**
12.	**a.**	Investigate the effect of CG location (Above chord line) on the longitudinal static stability of an aircraft.	L4	CO5	**10**
		OR			
	b.	Review the condition of lateral-directional static stability of a transport aircraft.	L4	CO5	**10**
13.	**a.**	Review the concept of direct operating cost (DOC) of typical jet airliner.	L4	CO6	**10**
		OR			
	b.	Examine the calculation method of direct operating cost (DOC) of typical jet airliner.	L4	CO6	**10**